

Optimization – 4

Discrete optimisation: rearrangement, exchange and greed

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Theoretical minimum.

Theorem 1 (rearrangement). For $a_1 \leq \dots \leq a_n$ and $b_1 \leq \dots \leq b_n$ the sum $\sum a_i b_{\sigma(i)}$ is largest for $\sigma = \text{id}$ and smallest for the reversing permutation. Chebyshev's sum inequality and Muirhead's inequality are corollaries; the proof is a single transposition together with Abel summation.

Theorem 2 (majorization). Write $x \prec y$ if the ordered partial sums of x are dominated by those of y and the totals agree. Then $x \prec y$ implies $\Phi(x) \leq \Phi(y)$ for every Schur-convex Φ , and Φ is Schur-convex as soon as it is symmetric and $(x_i - x_j)(\partial_i \Phi - \partial_j \Phi) \geq 0$.

Fact 1 (exchange, smoothing, greed). The universal discrete technique is: take an optimum, assume it violates the structure you expect, and exchange two elements to improve it. Smoothing is the same idea for continuous parameters. The greedy algorithm produces the optimum for every weight function exactly when the family of admissible sets is a matroid — this is the Rado–Edmonds theorem.

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1. Prove the rearrangement inequality twice, by a transposition argument and by Abel summation, and deduce Muirhead's inequality from it. Chebyshev's sum inequality, which is the other standard corollary, was proved in Limits – 7.
 2. (VJIMC 2002) The numbers $1, 2, \dots, n$ are assigned to the vertices of a regular n -gon in an arbitrary order. For each edge compute the product of the two numbers at its endpoints and sum these products. What is the smallest possible value of this sum?

3. (Proposed problems) For an integer $n \geq 1$ find the minimum of $f(\sigma) = \sum_{i \leq n/2} \sigma(i) + \sum_{i > n/2} \sigma^{-1}(i)$ over all permutations $\sigma \in S_n$, and exhibit a permutation realising it.
4. (CIIM 2010) On one side of a hall there are $2N$ rooms numbered 1 to $2N$; room i contains p_i beds. Every bed from the rooms $1, \dots, N$ must be moved to the rooms $N+1, \dots, 2N$ so that room j ends with p_j beds, where $p_1 + \dots + p_N = p_{N+1} + \dots + p_{2N}$. Moving a bed from room i to room j costs $(i-j)^2$. Find a moving plan of minimal total cost.
5. Prove the Rado–Edmonds theorem: the greedy algorithm returns a maximum weight independent set for every weight function if and only if the family of independent sets is a matroid. Check it on spanning trees and exhibit a non-matroid family where greed fails.